energies

MDPI

Review

Radiant conditioning retrofitting for residential buildings.

Hung Quang Do [1], Mark B. Luther [1,*], Mehdi Amirkhani [1], Zheng Wang [1], Igor Martek [1]

[1] School of Architecture and Built Environment, Deakin University, Geelong, VIC, Australia, 3220; qhdo@deakin.edu.au; m.amirkhani@deakin.edu.au (M.A.); igor.martek@deakin.edu.au, wangzheng@deakin.edu.au (Z.W.);

* Correspondence: mark.luther@deakin.edu.au

Abstract: In order to achieve Australia's greenhouse gas emissions reduction targets, a majority of existing residential building stock in Australia will require retrofitting in favour of energy-efficient solutions. The question arises as to what is the best retrofitting approach. In this paper, the range of retrofitting options are identified, and of these, conditioning retrofitting is assessed to be most straightforward and with the greatest potential to deliver significant positive results. Radiant conditioning systems are not new, yet some game-changing innovations have taken place over the last decade that may require an entire paradigm shift in fabricating buildings. The reiteration of the principle, 'thermally active systems', suggests that our buildings need to accommodate these systems into the fabric of building components. However, extremely little information in improving or innovative solutions for doing such seem to be provided by the industry. We seem incompetent with solutions that are not costing the earth, insulating, lightweight, and offering an instant response time to conditioning. We still have the concept embedded in our minds that radiative systems consist of heavy 'combat' construction with time lags of a day or two and that they are very costly to implement, especially if we are to retrofit a project. The purpose of this paper is to rectify and change our understanding of radiant systems, namely, through a review of the existing technology and its recent advancements. It intends to introduce the fact that radiant systems can become highly reactive, responsive, and thermally dynamic conditioning systems. Lightweight radiant systems can be 40% more energy-efficient than common air conditioners and can respond in less than 15 minutes, rather than in the hours required of heavy radiant systems. Thus, a lightweight radiant system is ideal for retrofitting residential buildings. Furthermore, this paper supports and introduces various systems suited to retrofitting a residential building with hydronic radiant systems.

Keywords: radiant system; conditioning system; residential building; retrofitting; thermal comfort

Citation: Lastname, F.; Lastname, F.; Lastname, F. Title. *Energies* **2021**, *14*, x. https://doi.org/10.3390/xxxxx

Academic Editor: Firstname Lastname

Received: date
Accepted: date
Published: date

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Copyright: © 2021 by the authors. Submitted for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction: retrofitting conditioning systems in residential buildings

1.1 The benefits of retrofitting condition systems

Australia has set a target of achieving a 26-28% reduction in greenhouse gas emissions by 2030 from 2005 levels and 25% of this targeted reduction needs to be achieved by improving energy efficiency in the building sector [1]. Meanwhile, Australia was slow in adopting energy-efficient building regulations leading to a huge number of housing projects that were built without any energy efficiency standards [1]. This leads to a gigantic need for retrofitting residential housing in Australia with new energy-efficient technologies. To bring forth maximum effectiveness, the retrofitting should be low cost and easy to be implemented [2]. Aside from meeting the national sustainable target, energy-efficient retrofitting housing can benefit occupants with lower energy costs and improved well-being. Ma et al [3] conducted a review of studies on the most modern methods of improving residential building thermal and energy performance. Among the studies on residential building retrofitting, envelope system refurbishment and mechan-

ical retrofitting prove most popular. Several attempts have also been made to improve the envelope of old residential housing in Australia. For example, Wilkinson and Feitosa [2] want to retrofit Australian housing with green roofs while Bulut et al [1] improved the windows with secondary glazing. Jamil et al [4] looked at the interior system and installation material used in ceilings to stabilize the thermal condition of residential housing. This research paper, however, considers a hitherto neglected approach to retrofitting: conditioning retrofitting.

In most states in Australia, air conditioners cause a serious problem to the electricity network as they are the main reason for peak energy consumption [5]. Energy usage of air conditioners is the biggest, making up 40% of the total energy consumption in Australian housing [5]. Meanwhile, even a small adjustment to the control systems of commonly used conditioning systems can result in significant savings in energy use and cost [6]. Also, while new housing projects in Australia have to meet a high standard of energy efficiency and benefit from new technologies, older residential housings built before 1980 do not [5]. This results in poorer energy performance especially for conditioning systems in older residential housings in Australia [5]. Additionally, Willand, Maller, and Ridley [7] conducted a study on the retrofitting of residential buildings in Melbourne and concluded that even small retrofits are capable of bringing forth notable improvements in residential building energy and thermal performance, reducing energy costs [7]. Hence, there is a need to retrofit the conditioning systems in Australian residential buildings, especially those built prior to 1980.

Over the past 100 years, convective systems have become the primary condition system in modern buildings worldwide with the development of mechanical ventilation. These systems in which air is used as the heat transfer medium for conditioning required spaces make up the Heating, Ventilating, and Air-conditioning (HVAC) industry [8]. While these conditioning systems can be successful for thermally heating and cooling, they are not necessarily the most economical and practical principle in producing thermal comfort [8, 9]. For example, decreasing the cooling temperature by 1°C will increase the cooling energy consumption by about 8-10% [10]. Miriel et al [11] argue that in the cooling mode, radiation accounts for about two-thirds of the heat transfer between the human body and room environment, while in the heating mode, 80% of total heat transfer is radiation. In short, radiant systems replace convection heat transferring between a human body and the surrounding environment with radiation. Additionally, many research papers have proven the merit of radiant systems over traditional HVAC in terms of energy and comparable thermal comfort performance [12]. These improved performance and advantages over HVAC have promoted radiant systems in most building sectors, including residential, commercial, education, and even industrial buildings of all sizes [13].

1.2. Comparison of energy consumption of radiant systems with existing HVAC systems

While mechanical conditioning systems have become fundamental in ensuring thermal comfort in modern buildings, conventional HVAC systems are highly energy-intensive [14]. The reason for this low performance is the inefficient heat transferring mechanism that the traditional air conditioners rely on [8]. Conventional air conditioning systems use air as the heat transferring medium and convection as the primary heat exchanging mode [13]. According to Moe [8], low-density air is a poor heat transferring medium, while only about 27% of human body heat transfer is via convection. Conventional HVAC systems are based on disadvantageous elements that create serious setbacks with regard to providing high thermal and energy performance. Additionally, air conditioning systems with large supply air volumes are also burdened by several limitations, such as noise, drafts, and poor temperature gradients [10, 14].

Moreover, the heat transferring mode of radiant systems has been shown to be superior. For example, Feustel and Stetiu [15], after running a numerical computer model, concluded that radiant cooling ceilings offer advantages over air conditioning systems despite their limited capacity. They suggested that by using water as a heat transferring medium, hydronic radiant systems can significantly reduce the amount of energy used in transporting heat and air, resulting in a 40% more energy-efficient system.

Imanari, Omori [16] conducted a study comprising a numerical simulation comparing annual energy consumption between two systems used by an office building, Tokyo, Japan. The result shows a 10% energy saving for the radiant ceiling. These saving means confirm that radiant systems in the context of office buildings prove to be financially viable, with the payback period being between 1 to 17 years [16]. The US department of technology has published three technical support documents recommending radiant conditioning systems as an alternative to HVAC, quoting a saving of up to 50% in energy consumption across small, medium, and large office buildings [12]. Hao et al [17], investigated the performance of radiant cooling ceilings, combined with displacement ventilation and desiccant dehumidification using simulation in the context of hot, humid summers, in Beijing, China. The radiant cooling system was compared to a conventional convective system, with results revealing a mere 8% total energy consumption. That is, the radiant system generates savings of up to 68.5% of chiller and 39.0% of fan energy systems. Hence, the whole radiant system proves to be more energy-efficient than the convective system, even when the radiant system consumes 25.6% more energy for pumping and about 4 times more for the boiler.

2. Research methodology

2.1. Literature review

This study is conducted based on a literature review of radiant systems. It confirms their potential as a preferred approach to conditioning retrofitting of residential buildings. As mentioned, retrofitting residential buildings is an emergent field of study [3]. However, most studies in this area were conducted using simulations or instrumental experiments. For example, Jamil et al [4] utilized simulations and instrumental experiments while the studies conducted by Bulut et al [1] and Wilkinson and Feitosa [2] were only conducted using instrumental experiments. Nevertheless, the knowledge in this field is rather well developed [3]. Radiant systems have also been comprehensively studied since the 1960s [18]. Moreover, these studies have been reviewed in several high-quality publications over the last six years. Despite that, the potential for radiant systems to service conditioning retrofitting of residential buildings has yet to be comprehensively appraised.

A review study was conducted by Rhee and Kim on radiant systems revealing 50 years of knowledge on radiant systems since the 1960s [18]. Rhee, Olesen, and Kim [19] and Karmann, Schiavon, and Bauman [12] also provide valuable overviews on the body of knowledge on radiant systems. The review studies on retrofitting residential housing of Ma et al [3] also have similar value categorizing the methods of housing retrofitting. Thus, a literature review serves as an appropriate initial approach in clarifying the application of radiant systems for the retrofitting of residential buildings. This literature review study can also lay a foundation for prototype development and further instrumental experiments.

2.2. Review method

This research paper was conducted mostly with qualitative data gathered from the previous publication since the 1960s. This is also the time used in the aforementioned high-quality review radiant systems paper of Rhee and Kim [18]. A breakthrough re-

search paper in radiant cooling was made in Australia in this period. The material was searched on google scholar linked with Deakin Library. The keywords applied in this research are "residential building energy consumption", "energy retrofitting residential buildings", "radiant conditioning", "radiant cooling and heating", "residential air conditioning systems", "radiant panel", "thermally active building systems", "mean radiant temperature". Several searches were made with Australian context or keywords such as "thermal comfort", "control", "measuring", "condensation prevention", "ventilation".

As many as 224 research papers were gathered and cataloged according to their research topics and content. Around 80 papers were marked as highly relevant containing valuable information and used in this study. Recently published papers and review papers were prioritized and used to trace back to earlier research papers. The REHVA handbook no 7, EN ISO 7730, and ASHRAE 55 2017 are critically important standards and guidelines for thermal comfort and conditioning principles as used in this research.

2.3. Findings

The data gathered from the previous studies are categorized and analyzed in order to shed light on the following salient concerns. What is the difference between new radiant systems and conditioning systems commonly used in residential buildings in Australia? By what mechanisms do radiant systems provide thermal comfort? What are the types of radiant systems and how well do they perform? What are their merits and limitations? Answering these questions allows a determination on the most appropriate evident configuration of radiant systems in the retrofitting of residential buildings.

3. Common conditioning systems in Australian houses

As mentioned above, air conditioners are the most commonly used systems providing thermal comfort for residential housing in various regions of the world such as China, USA, while the demand for air conditioners is dramatically increasing in China, India, Indonesia, and Brazil [20]. In Australia, the most frequently applied conditioning systems within the residential sector include the window-mounted or wall-mounted direct expansion A/C unit (Figure 1) [5]. These have traditionally provided both heating and air-conditioning (cooling) of a single room or zone within a house. Over the last two decades, these systems have been gradually replaced by split systems which consist of an external heat pump compressor and an internal wall-mounted fan (cassette) (Figure 2). The slipt system is also one of the most widely used conditioning systems in Australia [5]. Systems which are less popular in residential buildings in Australia are ducted (centralized conditioning systems), evaporative conditioners, and portable conditioners [5].

Figure 1: Wall-mounted direct expansion A/C. Source: First author

Figure 2: Split systems air conditioner. Source: First author

Other popular systems, used in heating only, are wall-mounted gas burners (Figure 3A) as well as completely ducted HVAC systems. It is not common to have both heating and cooling introduced within an HVAC ducted system. Further to these heating approaches, radiant panels, both hydronic and electric (Figure 4 B&C) are recently becoming popular. Hydronic systems often apply a gas or diesel-fueled boiler. It is important to note, that all of these conditioning systems primarily condition occupants under the principle of convection [21, 22].

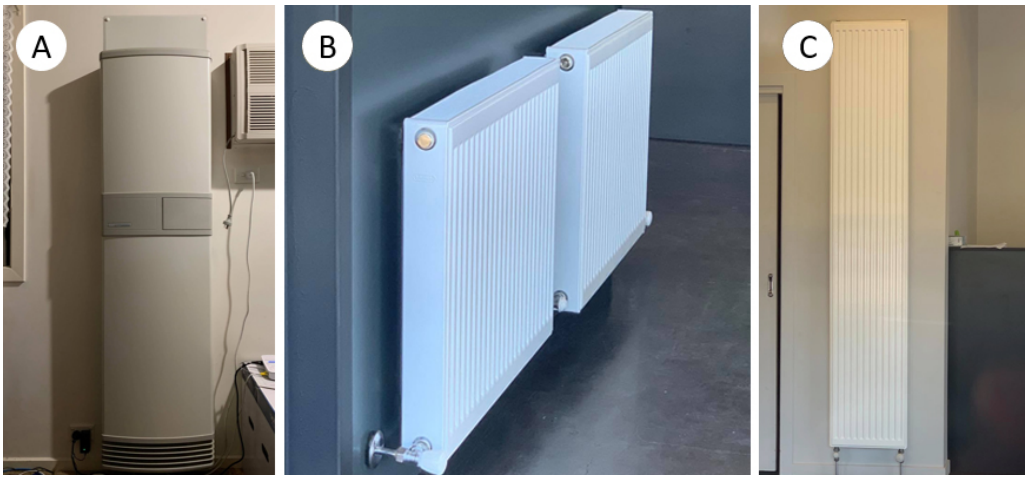


Figure 3: A) a convective heater; B) a hydronic radiator; C) a modern hydronic radiator panel. Source: First author

In the past, and again more recently, residential buildings in Australia have applied electric floor heating (Figure 4). These systems use a mat of electric heating cable that runs underneath the finished floor, turning the floor into a thermally active surface that radiates heat.

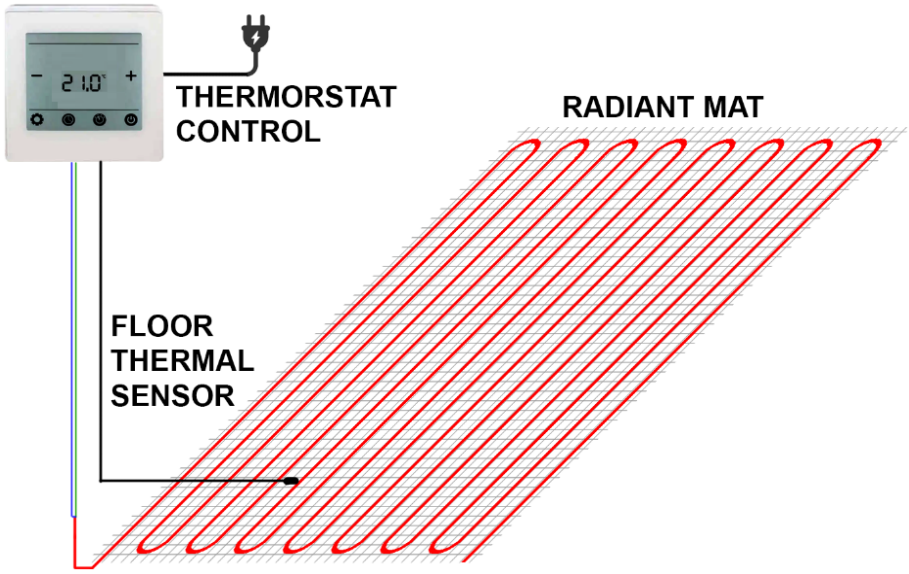


Figure 4: An electric floor heating system. Source: First author

A hydronic radiant mat is a recently introduced radiant system that mainly utilizes the radiant heating transferring mechanism, radiating heat directly to the occupants rather than conditioning the air temperature. This type of radiant system has more flexibility, simplifies installation, and localizes energy performance compared to HVAC systems or other types of radiators [23-25]. Radiant systems have become ubiquitous in North-East Asia and Europe [26]. While in the past, radiant systems are not yet widespread in North America, recently, the market for radiant systems in this region has shown rapid growth [26]. There has been no significant research on Australia's radiant systems in the last few decades [18]. In general, radiant systems can be considered superior to convective systems, both in terms of thermal and energy performance [12]. Hence, the prospect of replacing conventional HVAC with radiant systems could lead to better thermal comfort and reduce energy costs. As such, this research paper will investigate the capabilities of various types of radiant systems and the potential optimum application and control of hydronic radiant systems in retrofitting residential buildings. It also proposes an innovative hydronic radiant system based on the literature review. This research paper explains the principle of thermal comfort and the nature behind the benefit of radiant conditioning for housing project owners, conditioning systems manufacturers, and architects. Based on such knowledge, existing convective conditioning systems in residential housing can be retrofitted with radiant condition systems for better thermal and energy efficientcy results.

4. Thermal comfort

Thermal comfort can be defined as "the condition of the mind in which satisfaction is expressed with the thermal environment"[27]. However, several other definitions for thermal comfort have been pointed out, and even debates around how thermal comfort can be defined [28]. The problem is that the mechanism of being "thermally comfortable" is highly complicated and affected by various factors such as physiology, psychology, and behavior [29]. Hence, there are several comfort models used for different environments [30].

According to Fanger [31], thermal comfort can be quantitatively measured using six parameters, namely air velocity (Av), mean radiation temperature (MRT), air temperature (dry-bulb temperature – DB), air relative humidity (RH), metabolic rate (met), clothing insulation (clo). The combination of these parameters indicates the level of thermal comfort in a controlled and conditioned environment. The level of thermal

comfort (or discomfort) of a group of people is rated via predicted mean vote (PMV) index scaling from -3 (cold) to +3 (hot) [32]. Along with PMV, the index of the predicted percentage of dissatisfied (PPD) indicates the proportion of thermally dissatisfied people in the group. This principle makes up the classic PMV (Fanger) thermal comfort model shown in figure 5.

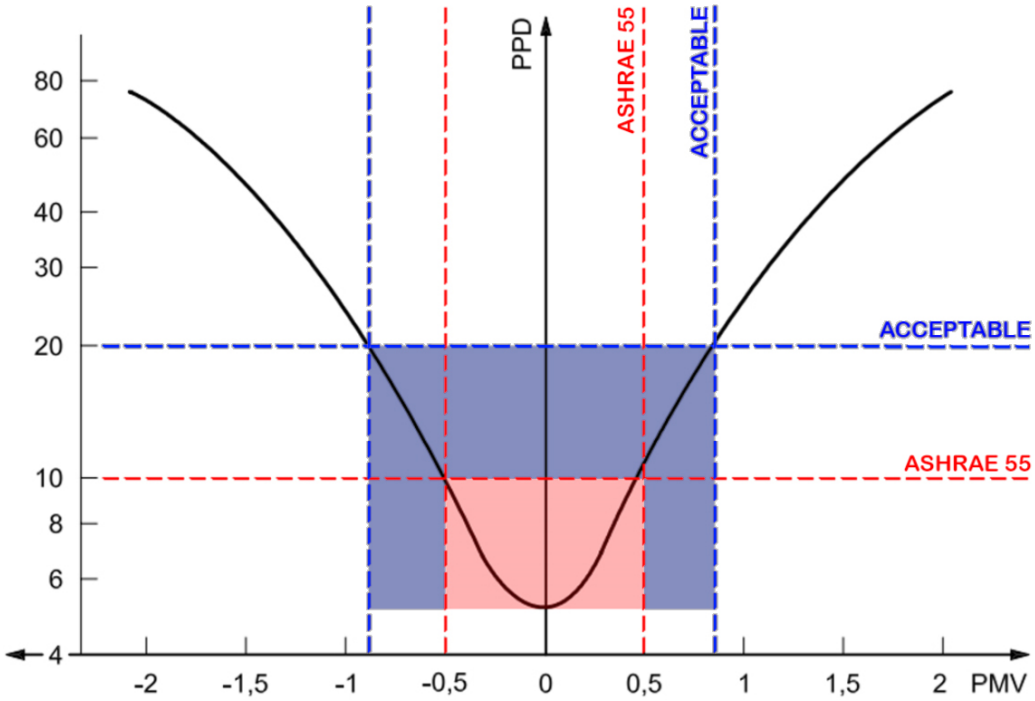


Figure 5: PMV (Fanger) thermal comfort models [27]

As the most classical thermal comfort model, the PMV thermal comfort model is commonly used in standards or guidelines such as ISO 7730 [32], ASHRAE 55 (standard of The American Society of Heating, Refrigerating and Air-Conditioning Engineers) [27], or the REHVA (Federation of European Heating, Ventilation, and Air Conditioning Associations) guide book number 7 [33]. The PMV thermal comfort model is well suited and commonly used to assess thermal comfort in conditioned office spaces [34]. Loveday et al [35] conducted an experiment proving that the PMV model accurately assesses thermal comfort in a radiant conditioned environment and concluded that a radiant conditioned space is similar to an air-conditioned space. According to ASHRAE 55, a thermal environment with an acceptable level of comfort would have a PMV between –0.5 and +0.5 resulting in a PPD of less than 10% (figure5) [27]. However, the Fanger PMV comfort model can be considered conservative [36], and acquiring a PPD of less than 10% is not common [27]. Hence, the PPD of less than 20% (PMV between –0.8 and +0.8) is also welcomed as an acceptable level of thermal comfort (Figure 5) [27, 36].

4.1. Mean radiant temperature (MRT)

MRT is a crucial parameter in determining thermal comfort [14, 27, 37]. It is defined as "the uniform surface temperature of an imaginary black enclosure in which the radiation from the occupant equals the radiant heat transfer in the actual non-uniform enclosure"[38]. Stefan Boltzmann defines MRT as "a temperature value to evaluate the total radiation flux over the human body"[39]. MRT is the relationship between indoor surface temperature, direct and scattering solar radiation power, view factor toward transparent and translucent surfaces, and solar radiation receiving area of the occupant body [40]. As such, MRT is the most complex parameter of thermal comfort, often difficult to control, calculate and measure [40, 41].

MRT is assumed to be equal to air temperature in many early research papers in space conditioning [38, 42, 43]. However, this is not the case on many occasions especially when solar heat gain is introduced [27, 44]. This is the condition of residential buildings. Instead, MRT is a better index to represent occupants' thermal sensation than air temperature [37, 45]. Hence, in retrofitting residential buildings, MRT control should be prioritized.

Sui and Zhang [10] investigated the relationship between MRT and other thermal elements, namely air temperature, air velocity, and air humidity. The research proved the influence of MRT on thermal comfort. Sui and Zhang [10] illustrate that thermal comfort can be achieved even at high air temperatures with a low MRT. This allows the concept of high air temperature cooling and low air temperature heating. Air temperature and MRT are the two most critical aspects determining thermal comfort [33]. Radiant conditioning focus on changing MRT via internal surface temperature rather than changing air temperature. Hence cooling can be achieved even in high air temperatures, while heating can be provided with low air temperature [33]. Additionally, Sui and Zhang [10] argue that while low air velocity means reductions in convection and evaporation, MRT can play a significant role to improve thermal comfort. Their research shows that the higher the humidity, the lower the MRT should be.

Figure 6 charts an investigation in achieving thermal comfort performed by Luther, Tokede, and Lui [36] applying similar principles to that of Sui and Zhang [10]. The circle and the triangle represent thermally unsatisfactory conditions, and there are three options to reach the acceptable level of thermal comfort (PPD=20%). Option A raises the air temperature by about 3.5°C, only which can increase conditioning energy cost by 30–45% [25]. Option B increases MRT by 4.5 °C to 23.5 °C (for the circle) and provides heating while the air temperature remains at 20.5 °C. This can be achieved with the radiant surface's temperature at 30°C–35°C at a minimum cost [25]. Meanwhile, HVAC systems need 8–10°C greater temperatures for heat to achieve the same thermal performance [25]. Hence, the radiant system is more energy-efficient. Nevertheless, option C raises both air temperature and MRT, the shortest path, although the energy consumption will be higher than option B.

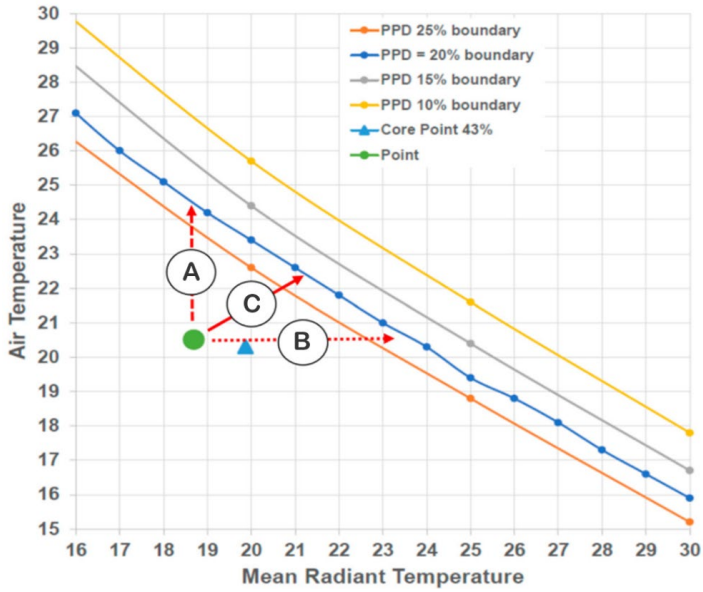


Figure **6**: Air temperature vs mean radiant temperature concerning PPD [25]

Additionally, providing a uniform MRT can prevent local thermal discomfort caused by air temperature gradients. Contrary to this, in acceptable vertical air temper-

ature gradients, the unfavorable difference in MRT can still cause thermal discomfort. In turn, with uniform MRT, local thermal discomfort can be avoided even with high air temperature gradients [35]. Hence, MRT has more effect on human thermal comfort than the air temperature in general [46].

MRT can be calculated using the surface's temperature and view factor (angle factor) between an occupant and internal surfaces. The equation is [47]:

$$T_r^4 = T_1^4 F_{p\text{-}1} + T_2^4 F_{p\text{-}2} + \ldots + T_N^4 F_{p\text{-}N}$$

Where

T_r = mean radiant temperature, K

T_N = surface temperature of surface N, K

Fp-N = view factor (angle factor) between a person and surface N

The view factor is the proportion of energy radiated from a surface and received by another surface [48]. The view factor represents the radiant heat exchanges between two surfaces [49]. In a room, the total view factor toward an occupant is equal to 1 [33]. The view factor is affected by the surface shapes, angle, and distance between the two surfaces[48]. Hence, in thermal comfort, the view factor depends on the dimensions of the radiant surfaces and the distance to the occupant. The view factor for a person in a room also depends on the posture and position [50]. Although determining the view factor is highly complicated, there are two conventional methods of approximately determining the view factor's value, namely the Fanger-Rizzo method and the Nusselt Analog method [51]. While the Fanger-Rizzo method is used in fundamental documents ASHRAE and REHVA handbook [33, 52], the Nusselt method is not. This is an important factor in designing radiant systems for retrofitting residential buildings, especially where the systems can be mounted for maximun effectiveness.

5. Hydronic radiant systems

A hydronic radiant system pumps hot or cold water through a series of pipes hidden inside or attached to the wall, floor, or ceiling [26] (figure 7). This will turn the wall, floor, or ceiling surfaces into "thermally active surfaces" for radiating heat or absorbing heat [8]. The radiant surfaces of the hydronic radiant system can also have a heat conduction effect on the air in contact with them and thus promote air convection.

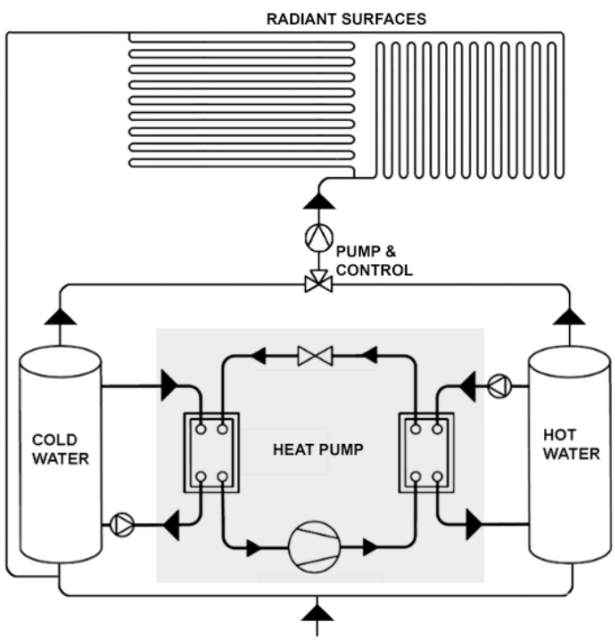


Figure 7. Radiant system diagram. Source: Fourth author

5.1. Hydronic radiant systems

The Federation of European Heating, Ventilation, and Air Conditioning Associations (REHVA) guidebook Number 7 classifies radiant conditioning systems into three types based on how the piping is installed [33], including:

1. Embedded surface system (ESS) with the tubing installed in a wall floor or ceiling layer but isolated from the main building structure,
2. Radiant panel (RP) where the tubing runs within prefabricated lightweight panels
3. TABS with the tube installed directly inside the building structure.

While the ESS and RP can be used in new constructions and retrofitting projects, TABS can only be used in new buildings. Two tube types are currently used in radiation systems, including small diameter tubes (metal or plastic tubes) and polyethylene capillary tube mats [53]. Hence, for more detailed classification, standard ISO-11855 listed seven types of radiant systems [23]. In this category, the ESS is divided into four subtypes depending on the tubing position (type A-C), type F with capillary tube, and type G with wooden structure, while type E is identical to TABS.

The new capillary mat system offers a promising retrofitting technology. This concept can be coupled together with an efficient heat pump as shown in Figure 7.

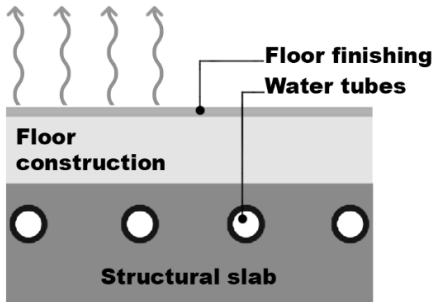
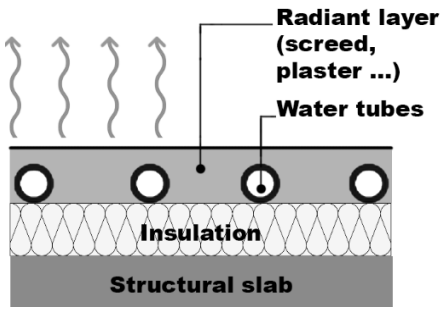
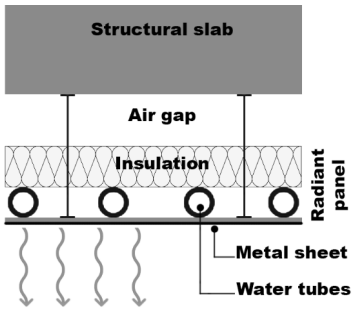
System types	Schematic	Classify
Thermal active building system (TABS)		Heavy-weight radiant system
Embedded surface system (ESS)		Medium-weight radiant system
Radiant panel (RP)		Light-weight radiant system

Table 1: Types of hydronic radiant systems [11] Source: First author

Table 1 illustrates three different radiant hydronic systems. TABS is the early radiant system that dates back to the 1930s [26]. It considers one substantial thermal mass that requires conditioning. This system can be relatively costly to run as it stores a great deal of energy within the building fabric and is not reactive to sudden climate changes. TABS also does not show a specific insulation layer, and the hydronic tubing is located deep inside the structure. This results in potential thermal and energy performance reductions. Several research papers have revealed the problems of heavy system design like TABS. As the sentinel load is stored and released by the thermal mass, the properties of the structure, such as thickness or thermal conductivity of the slab material, have significant impacts on the heat transfer process and thermal performance of this heavy system [54]. This heavy thermal mass results in slow react time, potential overheating, and low energy efficiency [18]. While the thermal mass can be advantageous in maintaining the temperature of radiant surfaces, receiving extra external load like solar heat gain can cause overheating or large temperature swings [55]. Heavy thermal mass can also reduce the efficiency of radiant conditioning systems as energy is wasted when space is not occupied or wasted on structure mass rather than contributing directly to the conditioning of the required environment [56]. In terms of reaction time, an experiment with radiant slab floor done by Merabtine et al [57] shows that the system surface needs about 4 hours to be heated up from 20°C to 29°C. Hence, this heavy system requires an advanced control system involving water flow rate, inlet water temperature, and reaction time [18]. Additionally, the slow reaction time can increase the risk of condensation in cooling mode if space's humidity increases faster than the system response [11].

In contrast to the TABS, the embedded surface system (ESS) has smaller thermal mass components. The radiant hydronic tubes are isolated from the main structure mass and located closer to the room surface. The ESS is a lighter system than TABS with less thermal mass resulting in lower thermal inertia [11], which means faster reaction time and less energy waste. The system is also isolated from the main structure. The insulation layer is crucial in directing the heat transfer [11], preventing energy waste.

Finally, RP is the lightest, with the radiant hydronic tube wholly isolated from any considerable thermal mass. RP systems are relatively new and improved systems that are lightweight, reactive systems [11]. The panels can be integrated into a ceiling system [58], which allows concentration on conditioning the environment with the radiant tube isolated from heavy thermal mass. Miriel et al [11] experimented with a radiant cooling ceiling panel with metal tubes resulting in the low responsive time, although the time is not explicitly stated. The commonly used type of radiant ceiling has the top surface insulated while the other downward surface is used for heat transfer [13]. The insulation layer on top of the system can have acoustic properties [11]. An experiment shows that a radiant panel system with tube mats has less than 15 minutes reaction times [53].

5.2. Radiant panel systems

Piping plays a critical role in RP performance. Pipe diameter, space between pipes, and thermal resistance of the pipe material significantly impact the surface temperature of radiant systems [54]. Besides water temperature, in heating, bigger pipes with low thermal resistance and less space between them result in higher surface temperature [54]. In short, more tube surface area and less material thermal resistance mean better surface temperature and more efficient heat transfer. There are two main piping, including small-bore tubes (about ø20 metal or plastic tube) and polyethylene capillary tube mats which are used for RP systems. Zhao et al [24] experiment on an embedded heating floor with ø20 plastic tube and a capillary mat gypsum heating panel system. The result shows that the capillary system can provide thermal comfort with inlet water of about 8°C lower and consume 45% less energy. According to Miriel, Serres [11], polyethylene capillary tube mats provide flexibility in design and installation. Additionally, the distance be-

tween tubes and heat resistance of tube and filling material in capillary tube radiant systems significantly influences the system's thermal capacity [25]. Capillary radiant systems are also considered more flexible and responsive than metallic tubes, although the capillary tubes can encounter blockage due to the small tube diameter [11].

Figure 8 shows a capillary radiant ceiling system is being installed in a residential project in Melbourne. The capillary mats are installed on the ceiling panels (area A), and the top with then be covered using insulation foam (area B). The upper insulation ensures the downward stream of heat transfer while the capillary tube enhances the thermal performance. This also proves that the flexibility and lightweight of polyethylene capillary tube mats allow installation with relative ease in various situations, making this type of piping ideal for retrofitting projects.

Figure 8: Capillary tube installation on a ceiling (KPW solutions Australia). Source: Second author

Figure 9 shows an EES installed on a radiant floor system in a residential project in Melbourne. The system's section is shown in Figure 10. The capillary tube mats are utilized to reduce the responding time that is negatively affected by the leveling mass and load-distributing layer needed for a solid floor system. Underneath the load-distributing layer is a solid insulation layer that prevents heat from reaching the structure mass directing heat to the top floor surface.

Figure 9: Capillary tube installation on a floor. Source: Second author

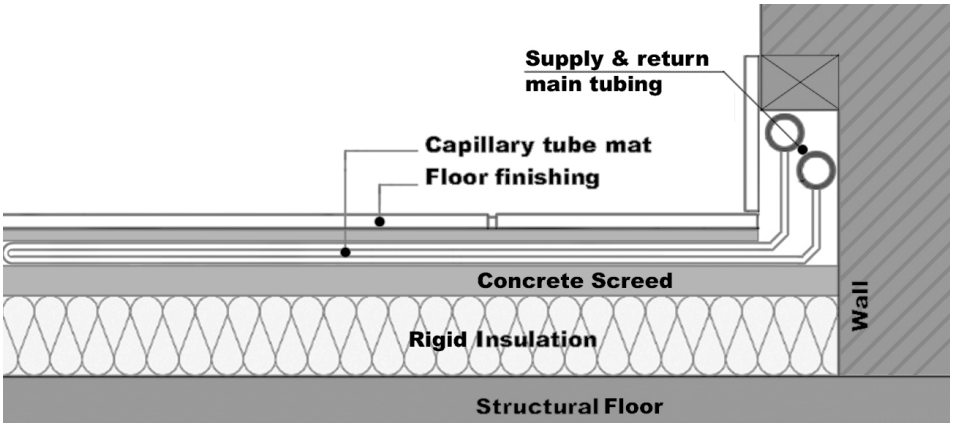
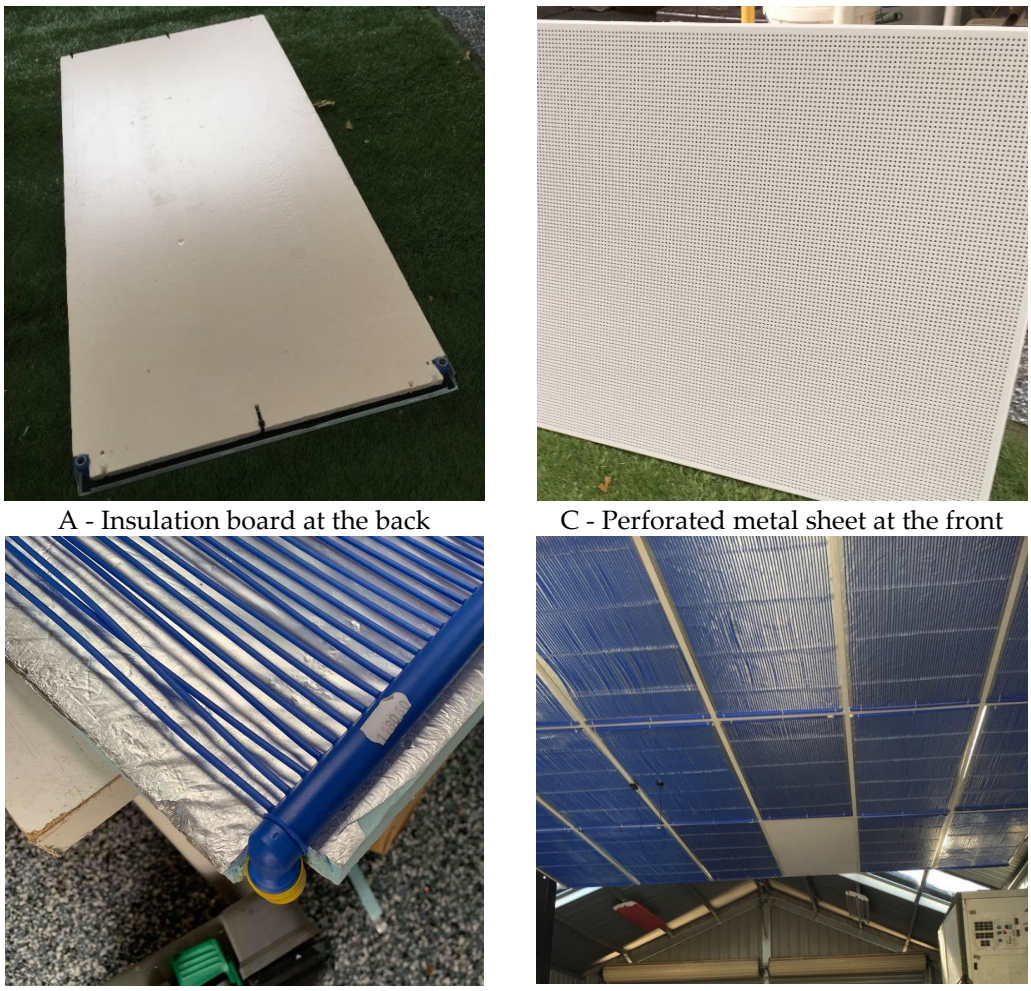


Figure 10: Capillary tube radiant floor. Source: First author

5.3. *A prototype for a lightweight and dynamic radiant cooling system*



A - Insulation board at the back

C - Perforated metal sheet at the front

B - An aluminum foil in contact tube mat

D – Integrated into a ceiling system

Figure 11: Capillary radiant panel system prototype.
Source: First and Second author

Based on the lessons learned from the previous review, we are proposing an RP system with polyethylene capillary tube mats (Figure 11). Each panel has a capillary tube mat attached to a solid insulation board (Figure 11A). The insulation board has a layer of aluminum foil in contact with the capillary tube to increase the heat directing effect (Figure 11B). The proposed finishing surface of a panel can be a perforated metal sheet or plasterboard that is in contact with the tube mat (Figure 11C). This finishing surface en-

sures the architectural appearance of the panel and the efficiency of radiant heat transfer. This lightweight and modular system allows integration to the ceiling system (11D) or wall cover. This prototype was tested, resulting in promising results. At present, this prototype is under testing, yet, indicating promising results.

The diagram of the proposed system is shown in figure 12. The design follows the principle diagram for the radiant system shown in figure 7 and includes several pumping solutions to ensure adequate water flow, reduce water leaks, and prevent blocking caused by air bubbles. The pressure tank is used to take the pressure changes in the system as the water temperature changes leading to changes in water volume. The vacuum relief valves are used to trap the bubbles and prevent blocking. At the same time, these valves can release pressure if the water pressure in the system increases to a critical level. A pressure gauge is used for monitoring the water pressure. The system functions correctly without leaking if the pressure is stable between 1-3 Bar (14.5-43.5 psi).

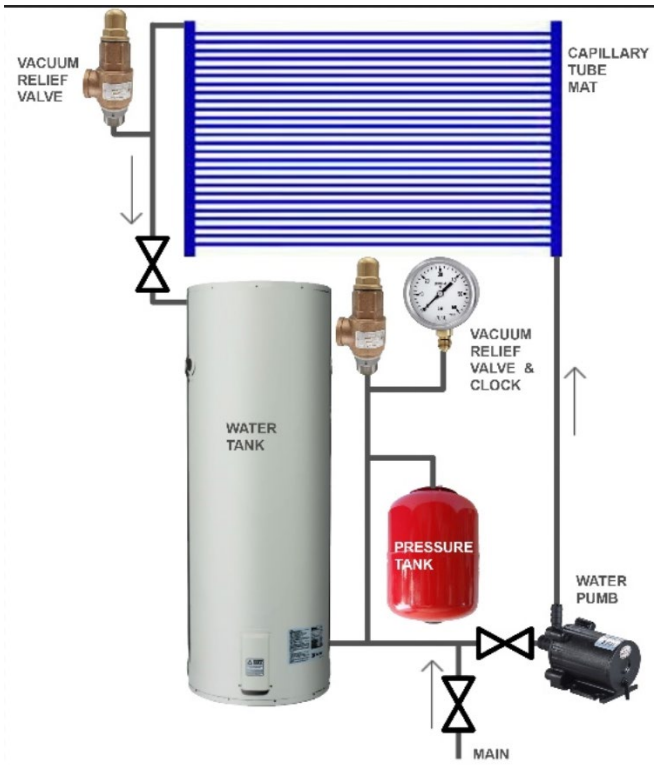


Figure 12: Capillary radiant panel system diagram. Source: First author

5.4. Mounting position for hydronic radiant systems

Figure 13 illustrates that a radiant cooling floor can provide the same level of thermal comfort as a radiant ceiling [10]. Olesen [59] explained that having a radiant cooling system mounted on the floor can be more advantageous than the ceiling due to the close distance between a floor and the occupants leading to a better view factor (angle factor). The angle factor is a fundamental aspect of radiant heat transfer representing the radiant heat exchanges between two surfaces [49]. A better view factor means more efficient heat transfer.

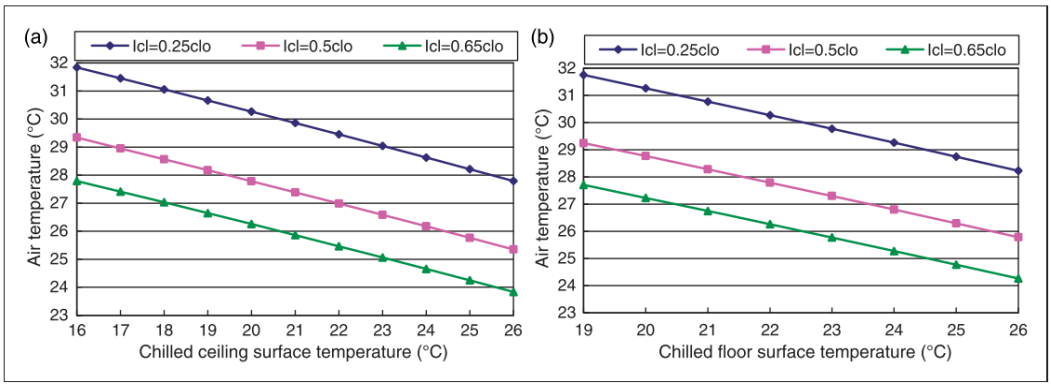


Figure 13. Thermal comfort between the radiant cooling ceiling (a) and floor (b) [10]

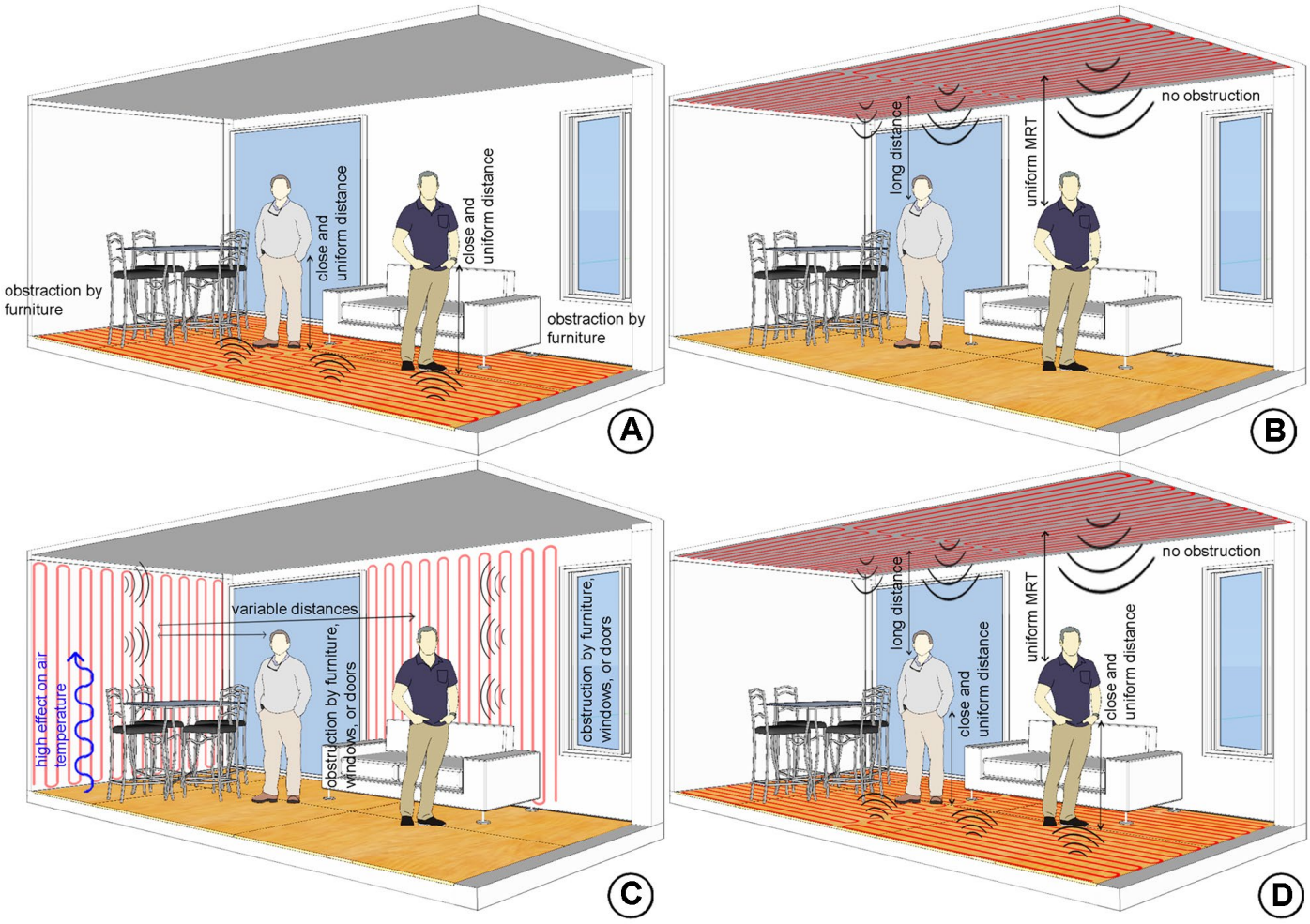


Figure 14: Floor-mounted radiant systems. Source: First author

Typically, the floor has the highest view factor compared to the wall and ceiling due to the close and uniform distance to the occupants (Figure 14A) [33], resulting in the best heat exchange coefficient. Hence the floor should be preferred. However, the efficiency of floor mounting radiant systems can be obstructed by furniture or carpet covering the floor radiant surfaces [60]. Fontana [61] experimented on the impact of furniture coverage on the thermal performance of the radiant heating floor. The result shows that furniture can significantly impact the heat flux between the radiant floor and the subject

environment, resulting in up to a 30% reduction in heat transfer. Hence furniture coverage must be taken into account when designing radiant systems [61].

Overall, the ceiling is the least favorable mounting location for radiant systems [58]. The reason can be the usual long-distance from the system to the occupants leading to low view factors. However, some clear advantages of the radiant ceiling can be seen, such as uniform MRT and usually not being covered (figure 14B).

Several research papers suggest that the floor is not the optimal mounting position for radiant systems. Bojić et al [58] compared the performance of radiant heating panels in 4 different positions. The results show that while mounting heating panels on the ceiling yield the worst thermal performance, heating panels on both floor and ceiling is the best option for thermal comfort (Figure 14D), followed by the heating wall and floor. Karabay, Arıcı, and Sandık [60] conducted a similar experiment comparing heating wall and floor with the result also favors the heating wall due to the better effect on room air temperature (Figure 14C) while not affected by furniture coverage. Having said that, while conducting a design for radiant walls, we found that radiant wall systems can still be affected by furniture coverage, while doors and windows can render radiant walls unfeasible on some occasions. In terms of view factor, as the distance between the radiant wall and occupants, can be various (Figure 14C) leading to un-uniform view factor and MRT. A question can be raised about to what extent, this un-uniform MRT can cause local thermal discomfort.

Meanwhile, both wall, ceiling, and floor mounting radiant systems can provide adequate thermal comfort. This proves the feasibility of the radiant system application. Although, in some buildings, it is not possible to install radiant panels at their preferred location, a radiant system with its flexibility can still be applied to other, less favorable locations and perform adequately.

6. Hydronic radiant system control

6.1. instrumentation for control

The study conducted by Sui and Zhang [10] shows the relationship between MRT and other measurable factors of thermal comfort and provides a basic mechanism in controlling the radiant systems. Firstly, air temperature is not a significant factor as a radiant system can provide cooling with high air temperature and hating low air temperature [33]. Secondly, the lower the MRT, the lower the air velocity should be to maintain thermal comfort. However, it is not practical to reduce the air velocity to 0 m/s. Thirdly, the lower the MRT, the higher the humidity should be to maintain thermal comfort [10]. However, high air humidity increases the risk of condensation which is the most problematic for radiant cooling systems. Meanwhile, the recommended humidity for TABS in cooling mode is only 40% [33]. Hence, the principle of controlling a radiant system can be summered as keeping the air velocity low, keeping the humidity in control, mainly adjusting MRT and adjusting air temperature if necessary.

The flow rate of water and the temperature of the inlet water significantly impact the hydronic radiant system thermal performance [62]. In particular, a steady and fast flow rate will ensure a uniform heat flux on the radiant surface and better thermal performance, while the water temperature is the most vital aspect ensuring the thermal and energy performance of the system [63]. Hence, to properly function, the water flow rate should be fast enough to provide a uniform temperature across the radiant surfaces, while the water temperature needs to be adjusted to maintain thermal comfort and control the energy consumption appropriately. There are two flow rate control mechanisms, namely on-off and variable control speed. However, flow rate control faces several disadvantages such as un-uniform radiant surface temperature, high fluctuations in room

temperature, and even noise make it far inferior to the more precise temperature control [64]. Nevertheless, the flow rate control method can be used in an individual room via a simple on-off control system linked with a room thermostat [59]. Commonly used TABS radiant systems operated with a stable flow rate while adjusting the water temperature [65].

As the water temperature is the most critical aspect in controlling radiant systems, most of the proposed control strategies focus on optimizing the water temperature supply to the subject spaces [64-67]. Olesen [68] pointed out that the set point for water temperature (average of inlet and return water) needs to be adjusted based on the outside temperature and specific building type regarding mass, heat loss, external and internal gain. The control mechanism recommended in the REHVA handbook for central control is based on the average water temperature (between the inlet and returned water) [33].

In hydronic systems, the flow rate and inlet water temperature are controlled by valves. For example, Miriel et al [11] used a three-way valve to control the inlet water temperature for cooling and heating while keeping the flow rate stable. The valve control is still the primary control for even modern hydronic radiant systems [18, 23, 33]. A simple control mechanism used in early radiant systems is adjusting the inlet water temperature and flow rate via manual valve control [33]. Radiant systems can also be controlled automatically using data provided by sensors [33]. These sensors collect the operative temperature data (MRT and air temperature) in a conditioned room and use the data on the control system and valves. Knowing the temperature of the inlet and return water, the mass flow rate, the heat transfer between the system and the conditioned space can be calculated and controlled, resulting in improved thermal and energy performance [33].

While the traditional thermostat systems control the conditioning systems using dry-bulb temperature the thermal monitoring method uses PMV index [69]. Meanwhile, controlling HVAC systems only with dry-bulb temperature is not adequate but PMV calculation should be used instead [70]. As the radiant conditioned spaces are similar to air-conditioned spaces in terms of thermal comfort [35], PMV calculation should also be the controlling index for radiant systems.

Modern thermal imaging technologies can potentially aid in automatic control for radiant systems. The infrared thermal cameras can provide detailed data on the room surfaces' temperature and heat sources [71]. The infrared thermal cameras allow data on MRT to be collected on a real-time basis and precise calculation of thermal comfort level (PMV and PPD) [69]. This infrared thermal imaging was used to provide data to the control systems of HVAC resulting in about 0.4 °C difference in conditioned air temperature in comparison with traditional thermostat systems [69]. This can be equal to 3.2 - 4% cooling load saving. Zampetti, Arnesano, and Revel [70] also test similar real-time thermal comfort monitoring systems to control heaters resulting in 17% energy saving and better thermal comfort. This thermal imaging technologies and thermal comfort monitoring methods should be tested with hydronic radiant systems.

6.2. Problem with thermal mass and controlling strategy

The main aim of an optimal control strategy for the radiant systems is achieving peak thermal and energy performance in operation [67]. This means maintaining an adequate level of thermal comfort when required while keeping energy consumption and operation costs low [33]. As some types of radiant systems are challenging to control and not reactive, control strategy plays a more critical role in ensuring performance [55]. In this case, radiant systems with high thermal mass proved to be the most challenging [23, 65]. The thermal mass can effectively reduce peak conditioning load and close the gap between day and night condition load [72]. However, this also leads to a slow responding

time that challenges system design and energy evaluation [73]. Therefore, high thermal mass radiant systems require efficient controlling strategies [65, 67, 72]. The REHVA handbook recommends using two controlling levels for high thermal mass radiant systems, namely central and individual control [33]. The center is used to set the water temperature and individual control to adjust the flow rate in each room [33]. On the other hand, Raftery et al [65] promote zone control alone as each thermal zone in a building may have different inlet water temperatures, average water temperatures, or flow rates.

Meanwhile, the radiant systems with low thermal mass, such as radiant panels, are more energy-efficient [33], more responsive, and easier to control; hence do not need a complex control strategy to properly function [23, 73]. This is a significant advantage of radiant panels over other high thermal mass radiant systems. Nevertheless, an improved control strategy is still a viable option for retrofitting existing high thermal mass systems.

The traditional control strategy for radiant systems is the continuous mode in which the heat flux between the system and the room is monitored, and the individual on-off control will respond accordingly to maintain the comfortable range of room temperature [74]. However, this strategy has drawbacks such as slow response to the external environment and high risk of discomfort [74]. Hence, advanced control strategies were developed considering the properties of high thermal mass systems [73]. Romaní, de Gracia, and Cabeza [73] mentioned some improved control strategies such as Pulse Width Modulation, Model Predictive Control, Adaptive and predictive controls, and Gain Scheduling Control. These advanced control strategies use data from weather forecasts or previous operations to set control variables and parameters [73].

7. Radiant cooling and condensation prevention

7.1. Radiant cooling

The idea of hydronic radiant cooling systems has been developed after heating and faces several difficulties. In the 1930s, Dr. Oscar Faber successfully apply a cooling capability to a radiant floor system installed in the Bank of England building [26]. However, the early hydronic chilled ceiling system developed in the 1930s was unsuccessful due to high condensation risk. Condensation remains one of the most significant problems of radiant cooling systems preventing radiant cooling from being widely utilized for several decades [75]. Condensation can also reduce the air quality of conditioned space and reduce the thermal efficiency of the radiant cooling systems [62]. Nevertheless, new knowledge and technology allow the problem to be avoided allowing radiant cooling systems to be constantly improved. Hence, recently, the interest in radiant cooling has returned, and radiant cooling systems have been used in several modern buildings even in humid climates such as Singapore or Thailand [19].

As mentioned above, condensation risk has been the biggest issue for radiant cooling systems from the very beginning in the 1930s [26]. Moreover, nearly 90 years later, Teitelbaum et al [75] suggested that the condensation risk caused by the low surface temperature of the radiant surface is one of the main reasons preventing radiant cooling systems from being commonly used, especially in hot, humid climates. Morse [76] pointed out that the dew point temperature for condensation is high in this type of climate, while the radiant cooling surface must be operated at a low temperature to balance the high air temperature and humidity. However, condensation will occur if the cooling surfaces' temperature is lower than the dew point. This makes setting up radiant cooling for hot, humid climates highly complicated. Additionally, condensation risk for cooling ceiling is significantly higher than cooling wall or floor [77] although cooling ceiling is the most common concept mentioned in radiant cooling research papers.

7.2. Preventing condensation

Many studies have been conducted to solve this condensation problem of radiant cooling, focusing on hot, humid climates. The common strategy is to combine radiant cooling with a mechanical ventilation system capable of dehumidification [17, 78-80]. This method aims to reduce the relative humidity and dew point temperature, allowing radiant cooling to be operated at the required temperature. Some research papers have been conducted to improve the efficiency of dehumidification ventilation. For example, Binghooth and Zainal [79], and Hao et al [17] use desiccant for dehumidification in ventilation systems to solve the problem in Malaysia's hot, humid climate conditions of Beijing, China. Both research papers concluded that condensation could be prevented along with energy-saving potential. Kim and Leibundgut [80] instead use hydraulic airbox units to reduce moisture in the supply air, ending with good results.

In a conditioned environment, air humidity mostly comes from air infiltration, and when using dehumidification ventilation, it is essential to give the system enough time to reduce the interior air humidity [78, 81]. Zhang and Niu [78] investigate the problem and recommend a strategy to prevent air infiltration and dehumidify the internal air at least one hour before the radiant cooling operation. Meanwhile, Ge, Xiao, and Wang [81] used a network of sensors and control systems to collect data from outdoor air quality and calculate the optimal radiant cooling strategy. The result is a significant reduction in dehumidification time to about half an hour.

Another approach in reducing condensation risk is to prevent condensation droplets from forming by using the property of the material surface. Yin, Wang [62] found out that radiant cooling systems with gypsum surfaces have less condensation risk than metal and plastic surfaces. Tang et al [82] looked at the condition of forming droplets on the surfaces of radiant cooling panels and propose super-hydrophobic treatment to the radiant surface of the cooling ceiling. The result is that the condensation droplets become too small to be noticed. However, this method can only make condensation neglectable rather than solving the problem.

Surprisingly, a breakthrough was made in the 1960s to tackle the condensation problem of radiant cooling in hot, humid climates [75]. Morse [76] proposed a radiant cooling system that can operate with the temperature below the dew point and make radiant cooling viable in hot humid. The principle is to isolate the cooling panel with a high humidity environment while allowing heat radiation to move almost freely. The author used a sheet of Polythene, a radiant transparency material, to cover the cooling surface and isolate it from the high humid air (figure 15). The result is a competent radiant system that has almost no impact on the air temperature or humidity. Hence, this system can be operated without any limitation on ventilation. Unfortunately, the principal did not receive much attention. After nearly half a century, Teitelbaum et al [75] apply this principle to build a radiant cooling system prototype and conduct an instrumental experiment. The author used polyethylene membrane as the cover sheet material (figure 16). The membrane allows more than 80% of the radiation to pass by, ensuring the system's thermal performance. The system was tested on an open bus stop in Singapore's hot, humid climate and proved that thermal comfort could be achieved even with a high metabolic rate and with no condensation risk. The result also shows that the membrane-assisted radiant cooling system can be operated at the surface temperature below the dew point and can even be used in natural ventilated environment.

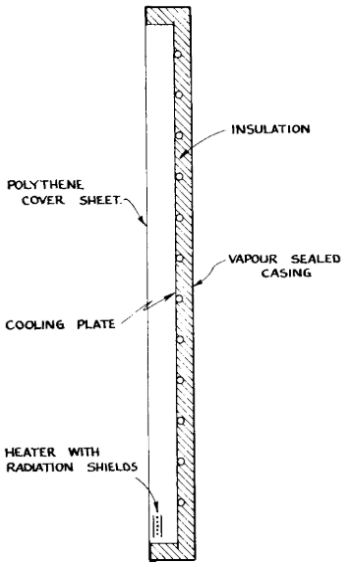


Figure 15: Morse [76] radiant absorbing panel.

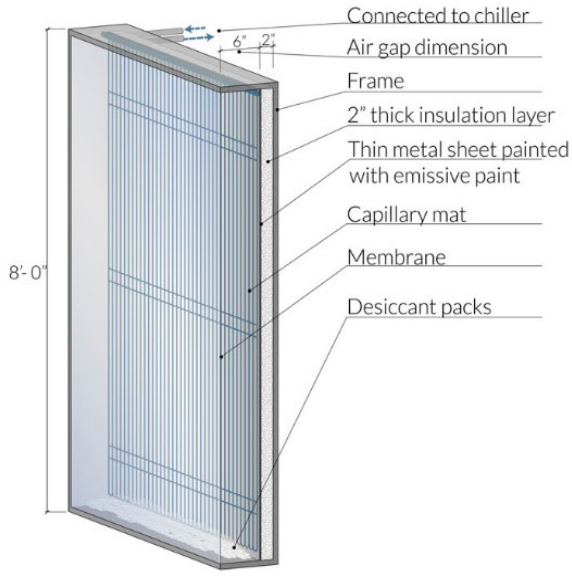


Figure 16: Teitelbaum, Chen [75] membrane assisted panel

7.3. Proposing a design for a condensation resisted radiant panel system

Figure 17 illustrates a proposed cooling radiant panel (RP) system inspired by the design of Morse [76]. The system utilizes capillary tube mats to reduce responding time and improve thermal performance. The rigid insulation boards can provide a heat direction effect as well as form the frame for the panel. The molded aluminum sheet behaves like a radiant plate and enhances the heat direction effect. The panel is sealed by a membrane (EPDM or PP or HDPE). This membrane allows most of the radiation from the environment to pass through and be absorbed by the tube mat and the aluminum sheet while isolating them from the environment. This design aims at providing a lightweight, dynamic and condensation resisted radiant panel system. This system is expected to be a solution for radiant cooling for retrofitting and new projects in hot humid climates. In further work, a prototype will be built and tested.

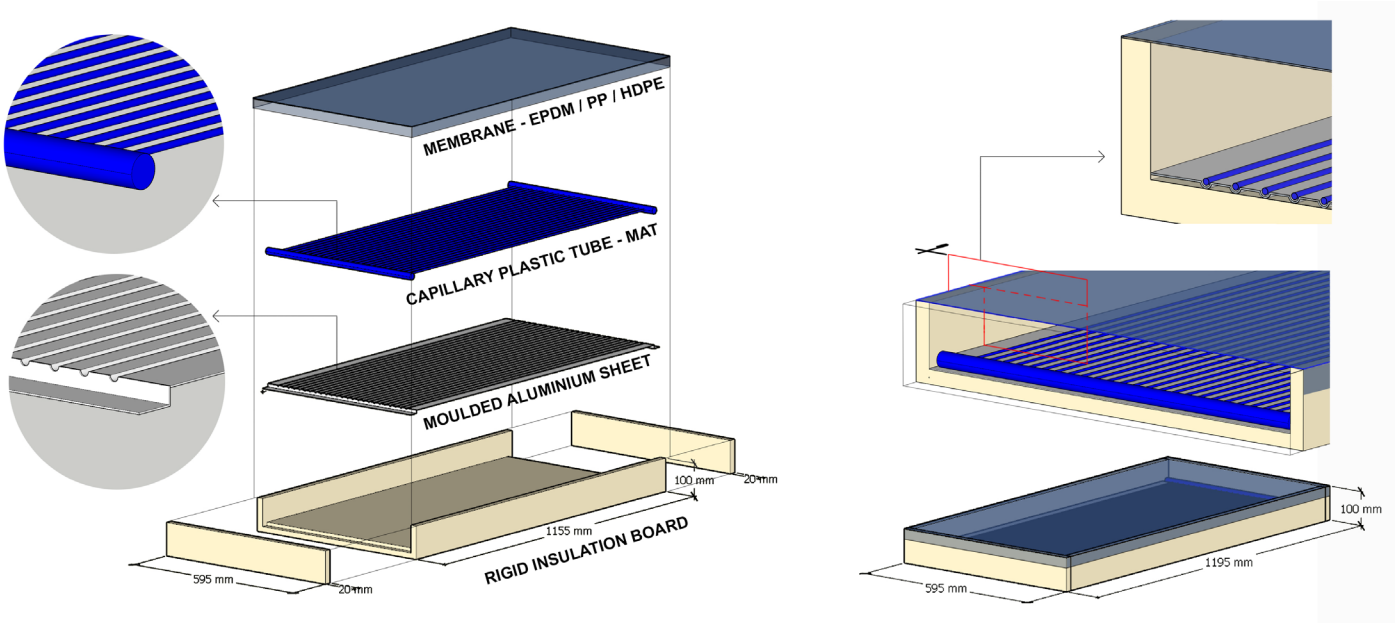


Figure 17: condensation resisted radiant cooling panel. Source: First author

8. Conclusion: Why retrofit residential housing with the radiant system?

8.1. General advantages of Radiant system

It is now well known among researchers those radiant systems offer high thermal performance, energy efficiency, and more [10, 18, 63, 68]. For example, Rhee, Olesen, and Kim [19] conducted a thoughtful literature review on the radiant system and concluded that hydronic radiant systems could offer several advantages: reducing the discomfort caused by the draft, low noise level, and good energy performance and design flexibility. The reduction in air velocity and risk of cold draft is the key aspect in radiant systems' superior thermal performance over HVAC systems [16, 83]. Specifically, radiant systems offer several advantages, including [10]:

- Creating a uniform thermal environment, reducing vertical temperature gradient and local discomfort.
- Requiring less air supply leads to a reduction in the discomfort caused by the draft.
- A decrease in the energy load for transporting air supply.
- Using water as a more efficient heat-carrying medium can significantly decrease the energy load for cooling transport.
- Incorporating MRT, which the human body senses the most resulting in more efficient energy performance.
- Cooling can be provided with relatively high water temperature; hence natural water resources can be used on some occasions.
- Quieter operation.

8.2. Limitation of radiant systems and recent improvements

Despite the merit mentioned earlier properties, radiant cooling systems still have several limitations. According to [16], the high price, difficulties in controlling, condensation risk, and system operational life are the issues of radiant systems. Similarly, Sui and Zhang [10] pointed out that problems, such as Indoor thermal environment design, system control, and anti-condensation, need to be solved for radiant cooling systems in residential buildings. Later, Serageldin et al [13] repeated some of the main issues with hydronic: condensation risk, difficulty in control, cold draught, high initial cost, and small capacity. Also, local discomfort can occur in a radiant heating environment caused by the distance between the human body and the radiant surface [18].

There is also complexity in installing and utilizing radiant systems. As the radiant system alone cannot supply fresh air to the conditioned space, there must always be an air supply system [84]. This, however, may not be the case for the leaky windows of old Australian residential buildings where air can be naturally circulated. This contributes to implementing radiant systems and hinders radiant systems as secondary conditioning options [63]. The pumping water system required for hydronic radiant systems can also be considered complicated, contributing to installation [85]. Another problem with the radiant floor systems is that radiant heat transfer is affected by furniture coverage [61] and carpet [60].

8.4. The potential solution of lightweight capillary tube radiant systems

The above review also highlights the advantage of lightweight radiant panel systems over heavier systems such as TABS or ESS. In general, lightweight radiant systems are:

- do not require complicated control systems.
- more flexible and easier to install than their heavier counterpart.
- more dynamic, and respond faster.

The thermal dynamic and fast response time of radiant panel systems can be further enhanced with capillary tube mats.

These properties along with the aforementioned advantages of radiant systems in general make the radiant panel systems ideal for retrofitting residential buildings and even other types of buildings such as commercial or industrial. However, further studies are needed to solidly confirm this. Hence an instrumental experiment is being developed.

References

1. Bulut, M.B., et al., *Thermal performance of retrofitted secondary glazed windows in residential buildings–two cases from Australia.* Smart and Sustainable Built Environment, 2021.
2. Wilkinson, S. and R.C. Feitosa, *Retrofitting housing with lightweight green roof technology in Sydney, Australia, and Rio de Janeiro, Brazil.* Sustainability, 2015. **7**(1): p. 1081-1098.
3. Ma, Z., et al., *Existing building retrofits: Methodology and state-of-the-art.* Energy and buildings, 2012. **55**: p. 889-902.
4. Jamil, H., et al., *Investigation of PCM as retrofitting option to enhance occupant thermal comfort in a modern residential building.* Energy and Buildings, 2016. **133**: p. 217-229.
5. Goldsworthy, M., *Towards a residential air-conditioner usage model for Australia.* Energies, 2017. **10**(9): p. 1256.
6. Simshauser, P. and T. Nelson, *The outlook for residential electricity prices in Australia's National Electricity Market in 2020.* The Electricity Journal, 2013. **26**(4): p. 66-83.
7. Willand, N., C. Maller, and I. Ridley, *Addressing health and equity in residential low carbon transitions–Insights from a pragmatic retrofit evaluation in Australia.* Energy Research & Social Science, 2019. **53**: p. 68-84.
8. Moe, K., *Thermally Active Surfaces in Architecture*. 2010: Princeton Architectural Press.
9. de Dear, R. and F. Zhang. *Why the perfect office temperature is a myth*. 2019 [cited 2021 20/07/2021]; Available from: https://www.sydney.edu.au/news-opinion/news/2019/03/14/why-the-perfect-office-temperature-is-a-myth.html.
10. Sui, X. and X. Zhang, *Analysis on combinations of indoor thermal microclimate parameters in radiant cooled residential buildings and drawing of new thermal comfort charts.* Building Services Engineering Research & Technology, 2016. **37**(1): p. 66-84.
11. Miriel, J., L. Serres, and A. Trombe, *Radiant ceiling panel heating–cooling systems: experimental and simulated study of the performances, thermal comfort and energy consumptions.* Applied Thermal Engineering, 2002. **22**(16): p. 1861-1873.
12. Karmann, C., S. Schiavon, and F. Bauman, *Thermal comfort in buildings using radiant vs. all-air systems: A critical literature review.* Building and Environment, 2017. **111**: p. 123-131.
13. Serageldin, A.A., et al., *Numerical investigation of the thermal performance of a radiant ceiling cooling panel with segmented concave surfaces.* Journal of Building Engineering, 2021. **42**: p. 102450.
14. Catalina, T., J. Virgone, and F. Kuznik, *Evaluation of thermal comfort using combined CFD and experimentation study in a test room equipped with a cooling ceiling.* Building and Environment, 2009. **44**(8): p. 1740-1750.
15. Feustel, H.E. and C. Stetiu, *Hydronic radiant cooling—preliminary assessment.* Energy and buildings, 1995. **22**(3): p. 193-205.
16. Imanari, T., T. Omori, and K. Bogaki, *Thermal comfort and energy consumption of the radiant ceiling panel system.: Comparison with the conventional all-air system.* Energy and buildings, 1999. **30**(2): p. 167-175.
17. Hao, X., et al., *A combined system of chilled ceiling, displacement ventilation and desiccant dehumidification.* Building and Environment, 2007. **42**(9): p. 3298-3308.
18. Rhee, K.-N. and K.W. Kim, *A 50 year review of basic and applied research in radiant heating and cooling systems for the built environment.* Building and Environment, 2015. **91**: p. 166-190.
19. Rhee, K.-N., B.W. Olesen, and K.W. Kim, *Ten questions about radiant heating and cooling systems.* Building and Environment, 2017. **112**: p. 367-381.

20. Debnath, K.B., et al., *Understanding residential occupant cooling behaviour through electricity consumption in warm-humid climate.* Buildings, 2020. **10**(4): p. 78.
21. Pedersen, T.H., et al., *The effect of including hydronic radiator dynamics in model predictive control of space heating.* Energy and Buildings, 2019. **183**: p. 772-784.
22. Embaye, M., R. Al-Dadah, and S. Mahmoud, *Thermal performance of hydronic radiator with flow pulsation–Numerical investigation.* Applied Thermal Engineering, 2015. **80**: p. 109-117.
23. Feng, J.D., *Design and control of hydronic radiant cooling systems*, in *Dept. of Architecture*. 2014, University of California.
24. Zhao, M., et al., *Performance comparison of capillary mat radiant and floor radiant heating systems assisted by an air source heat pump in a residential building.* Indoor and Built Environment, 2017. **26**(9): p. 1292-1304.
25. Ding, P., et al., *Study on heating capacity and heat loss of capillary radiant floor heating systems.* Applied Thermal Engineering, 2020. **165**: p. 114618.
26. Bean, R., B.W. Olesen, and K.W. Kim, *Part 2: history of radiant heating & cooling systems.* Ashrae Journal, 2010. **52**(2): p. 50.
27. ASHRAE, *ASHRAE Standard 55-2017: Thermal environment conditions for human occupancy*. 2017, Atlanta, GA: American Society of Heating, Refrigerating and Air Conditioning Engineers Inc.
28. Bessoudo, M., et al., *Indoor thermal environmental conditions near glazed facades with shading devices–Part I: Experiments and building thermal model.* Building and Environment, 2010. **45**(11): p. 2506-2516.
29. Anderson, T. and M. Luther, *Designing for thermal comfort near a glazed exterior wall.* Architectural Science Review, 2012. **55**(3): p. 186-195.
30. Zhao, Q., Z. Lian, and D. Lai, *Thermal Comfort models and their developments: A review.* Energy and Built Environment, 2020.
31. Fanger, P.O., *Assessment of man's thermal comfort in practice.* Occupational and Environmental Medicine, 1973. **30**(4): p. 313-324.
32. Standardization, I.O.f., *ISO 7730 2005-11-15 Ergonomics of the Thermal Environment: Analytical Determination and Interpretation of Thermal Comfort Using Calculation of the PMV and PPD Indices and Local Thermal Comfort Criteria*. 2005: ISO.
33. Babiak, J., B.W. Olesen, and D. Petras, *Low temperature heating and high temperature cooling: REHVA GUIDEBOOK No 7.* 2007.
34. Fanger, P.O. and J. Toftum, *Extension of the PMV model to non-air-conditioned buildings in warm climates.* Energy and buildings, 2002. **34**(6): p. 533-536.
35. Loveday, D., et al., *Displacement ventilation environments with chilled ceilings: thermal comfort design within the context of the BS EN ISO7730 versus adaptive debate.* Energy and Buildings, 2002. **34**(6): p. 573-579.
36. Luther, M., O. Tokede, and C. Liu, *Applying a comfort model to building performance analysis.* Architectural Science Review, 2020. **63**(6): p. 481-493.
37. Hua Ge, P., *An infrared sphere method to measure mean radiant temperature.* ASHRAE Transactions, 2013. **119**: p. 1F.
38. Chung, J.D., H. Hong, and H. Yoo, *Analysis on the impact of mean radiant temperature for the thermal comfort of underfloor air distribution systems.* Energy and Buildings, 2010. **42**(12): p. 2353-2359.
39. Chen, Y.-C., T.-P. Lin, and A. Matzarakis, *Comparison of mean radiant temperature from field experiment and modelling: a case study in Freiburg, Germany.* Theoretical and applied climatology, 2014. **118**(3): p. 535-551.
40. La Gennusa, M., et al., *The calculation of the mean radiant temperature of a subject exposed to the solar radiation—a generalised algorithm.* Building and Environment, 2005. **40**(3): p. 367-375.
41. Kántor, N. and J. Unger, *The most problematic variable in the course of human-biometeorological comfort assessment—the mean radiant temperature.* Central European Journal of Geosciences, 2011. **3**(1): p. 90-100.
42. Walikewitz, N., et al., *The difference between the mean radiant temperature and the air temperature within indoor environments: A case study during summer conditions.* Building and Environment, 2015. **84**: p. 151-161.

43. Chaudhuri, T., et al. *On assuming Mean Radiant Temperature equal to air temperature during PMV-based thermal comfort study in air-conditioned buildings*. in *IECON 2016-42nd Annual Conference of the IEEE Industrial Electronics Society*. 2016. IEEE.
44. Chaiyapinunt, S., et al., *Performance rating of glass windows and glass windows with films in aspect of thermal comfort and heat transmission.* Energy and Buildings, 2005. **37**(7): p. 725-738.
45. Alfano, F.R.d.A., et al., *On the measurement of the mean radiant temperature and its influence on the indoor thermal environment assessment.* Building and Environment, 2013. **63**: p. 79-88.
46. Barnaby, C.S. and C.O. Pedersen, *Develop a Radiant System Module for the Simulation and Analysis of Spaces and Systems.* ASHRAE Transactions, 2015. **121**(2).
47. Handbook-Fundamentals, A., *American society of Heating.* Refrigerating and Air-Conditioning Engineers, 2009.
48. Vujičić, M., N. Lavery, and S. Brown, *View factor calculation using the Monte Carlo method and numerical sensitivity.* Communications in numerical methods in Engineering, 2006. **22**(3): p. 197-203.
49. Lee, D.-S., et al., *A field study on application of infrared thermography for estimating mean radiant temperatures in large stadiums.* Energy and Buildings, 2019. **202**: p. 109360.
50. Olesen, B.W., *Thermal comfort*. Vol. 2. 1982: Bruel & Kjaer.
51. Guo, H., et al., *On the understanding of the mean radiant temperature within both the indoor and outdoor environment, a critical review.* Renewable and Sustainable Energy Reviews, 2020. **117**: p. 109207.
52. ASHREA, *ASHREA Handbook Fundamentals*. SI ed. Ashrae Handbook Fundamentals. 2021, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc: Atlanta GA
53. Díaz, N.F. and C. Cuevas, *Testing and thermal modeling of radiant panels systems as commissioning tool.* Energy conversion and management, 2010. **51**(12): p. 2663-2677.
54. Liu, Y., D. Wang, and J. Liu, *Study on heat transfer process for in-slab heating floor.* Building and Environment, 2012. **54**: p. 77-85.
55. Athienitis, A.K., *Investigation of thermal performance of a passive solar building with floor radiant heating.* Solar energy, 1997. **61**(5): p. 337-345.
56. Wang, Z., et al., *A model to compare convective and radiant heating systems for intermittent space heating.* Applied Energy, 2018. **215**: p. 211-226.
57. Merabtine, A., et al., *Experimental and multidimensional numerical analysis of the thermal behavior of an anhydrite radiant slab floor heating system: A multi-objective sensitivity study.* Energy and Buildings, 2018. **174**: p. 619-634.
58. Bojić, M., et al., *Performances of low temperature radiant heating systems.* Energy and Buildings, 2013. **61**: p. 233-238.
59. Olesen, B., *Radiant floor cooling systems.* Ashrae Journal, 2008. **50**(9): p. 16-22.
60. Karabay, H., M. Arıcı, and M. Sandık, *A numerical investigation of fluid flow and heat transfer inside a room for floor heating and wall heating systems.* Energy and Buildings, 2013. **67**: p. 471-478.
61. Fontana, L., *Thermal performance of radiant heating floors in furnished enclosed spaces.* Applied Thermal Engineering, 2011. **31**(10): p. 1547-1555.
62. Yin, Y., et al., *Experimental investigation on the heat transfer performance and water condensation phenomenon of radiant cooling panels.* Building and environment, 2014. **71**: p. 15-23.
63. Tye-Gingras, M. and L. Gosselin, *Comfort and energy consumption of hydronic heating radiant ceilings and walls based on CFD analysis.* Building and Environment, 2012. **54**: p. 1-13.
64. Ryu, S.-R., et al., *A study on the control methods for radiant floor heating and cooling system in residential building.* ASHRAE Transactions, 2004. **110**: p. 106.
65. Raftery, P., et al., *A new control strategy for high thermal mass radiant systems.* 2017.

66. Karlsson, H. and C.-E. Hagentoft, *Application of model based predictive control for water-based floor heating in low energy residential buildings.* Building and environment, 2011. **46**(3): p. 556-569.
67. Joe, J. and P. Karava, *A model predictive control strategy to optimize the performance of radiant floor heating and cooling systems in office buildings.* Applied Energy, 2019. **245**: p. 65-77.
68. Olesen, B.W., *Radiant floor heating in theory and practice.* ASHRAE journal, 2002. **44**(7): p. 19-26.
69. Revel, G., M. Arnesano, and F. Pietroni. *An innovative low cost IR system for real-time measurement of human thermal comfort*. in *Proceedings of IAQ2013 conference, Vancouver (Canada)*. 2013.
70. Zampetti, L., M. Arnesano, and G. Revel, *Experimental testing of a system for the energy-efficient sub-zonal heating management in indoor environments based on PMV.* Energy and Buildings, 2018. **166**: p. 229-238.
71. Revel, G., E. Sabbatini, and M. Arnesano, *Development and experimental evaluation of a thermography measurement system for real-time monitoring of comfort and heat rate exchange in the built environment.* Measurement Science and Technology, 2012. **23**(3): p. 035005.
72. Lehmann, B., V. Dorer, and M. Koschenz, *Application range of thermally activated building systems tabs.* Energy and buildings, 2007. **39**(5): p. 593-598.
73. Romaní, J., A. de Gracia, and L.F. Cabeza, *Simulation and control of thermally activated building systems (TABS).* Energy and Buildings, 2016. **127**: p. 22-42.
74. Cho, S. and M. Zaheer-Uddin, *Predictive control of intermittently operated radiant floor heating systems.* Energy Conversion and Management, 2003. **44**(8): p. 1333-1342.
75. Teitelbaum, E., et al., *Membrane-assisted radiant cooling for expanding thermal comfort zones globally without air conditioning.* Proceedings of the National Academy of Sciences, 2020. **117**(35): p. 21162-21169.
76. Morse, R., *Radiant cooling.* Architectural Science Review, 1963. **6**(2): p. 50-53.
77. Tang, H., X.-H. Liu, and Y. Jiang, *Theoretical and experimental study of condensation rates on radiant cooling surfaces in humid air.* Building and Environment, 2016. **97**: p. 1-10.
78. Zhang, L. and J. Niu, *Indoor humidity behaviors associated with decoupled cooling in hot and humid climates.* Building and Environment, 2003. **38**(1): p. 99-107.
79. Binghooth, A. and Z. Zainal, *Performance of desiccant dehumidification with hydronic radiant cooling system in hot humid climates.* Energy and Buildings, 2012. **51**: p. 1-5.
80. Kim, M.K. and H. Leibundgut, *Advanced Airbox cooling and dehumidification system connected with a chilled ceiling panel in series adapted to hot and humid climates.* Energy and buildings, 2014. **85**: p. 72-78.
81. Ge, G., F. Xiao, and S. Wang, *Neural network based prediction method for preventing condensation in chilled ceiling systems.* Energy and Buildings, 2012. **45**: p. 290-298.
82. Tang, H., et al., *Study on the reduction of condensation risks on the radiant cooling ceiling with superhydrophobic treatment.* Building and Environment, 2016. **100**: p. 135-144.
83. Kitagawa, K., et al., *Effect of humidity and small air movement on thermal comfort under a radiant cooling ceiling by subjective experiments.* Energy and buildings, 1999. **30**(2): p. 185-193.
84. Loveday, D.L., et al., *Designing for thermal comfort in combined chilled ceiling/displacement ventilation environments/Discussion.* ASHRAE transactions, 1998. **104**: p. 901.
85. Peng, P., et al., *Investigation on thermal comfort of air carrying energy radiant air-conditioning system in south-central China.* Energy and Buildings, 2019. **182**: p. 51-60.